

EFFICIENT BAYESIAN INFERENCE FROM NOISY PAIRWISE COMPARISONS

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ABSTRACT

The evaluation of generative models remains technically difficult because conventional automatic metrics frequently fail to approximate human preference judgments. Human evaluation provides a more direct measurement signal, but the resulting observations are expensive to collect and statistically noisy because raters differ in expertise, attention, and diligence. Pairwise-comparison protocols reduce some response variability, but converting sparse and noisy binary preferences into calibrated global quality scores requires an explicit probabilistic aggregation model. Bradley-Terry-based estimators update item scores from comparison outcomes, yet existing variants either omit rater-level heterogeneity or rely on optimization procedures without convergence guarantees, thereby limiting robustness and interpretability. We introduce BBQ, a Bayesian Bradley-Terry model that parameterizes rater quality, attenuates or removes the contribution of unreliable participants, and admits an Expectation-Maximization algorithm with guaranteed monotonic improvement of the likelihood. Across empirical evaluations, BBQ exhibits faster convergence, calibrated uncertainty estimates, and more stable and interpretable rankings than baseline Bradley-Terry models, including under noisy and crowdsourced annotation regimes. The resulting framework supports more reliable and cost-efficient human evaluation pipelines for generative models.

1 INTRODUCTION

Evaluating generative models is technically challenging, especially for large language models (LLMs) and image generators, because standard automatic metrics often provide weak proxies for human preference. Metrics such as BLEU (Papineni et al., 2002) and perplexity (Jelinek, 1998) for LLMs, or PSNR (Gonzalez, 2009), MS-SSIM (Wang et al., 2003), and FID (Heusel et al., 2017) for image models, characterize only restricted aspects of perceived quality (Mentzer et al., 2020; CLIC, 2025; Chiang et al., 2024). Consequently, human evaluation remains necessary when the target quantity is a meaningful ordering among competing models. Leaderboards and comparative studies for large language models and Learned Image Compression (LIC) systems therefore rely heavily on preference data collected from human raters.

Human evaluation is, however, costly in both time and annotation budget, so the evaluation protocol must maximize statistical efficiency while controlling subjectivity and measurement noise. Pairwise comparison protocols address part of this requirement by asking a participant to select the preferred item from a pair rather than assigning an absolute score (Zerman et al., 2018; Wang et al., 2023). This binary-choice interface is typically easier for raters and yields more reproducible judgments. Exhaustive pairwise collection is nevertheless impractical because the number of possible pairs scales quadratically with the number of items. Under a sparse comparison design, raw win rates are not valid global quality estimates, since each observed win rate is confounded by the strength of the specific opponents against which the item was evaluated.

Producing a leaderboard-level ordering therefore requires an aggregation procedure that maps sparse pairwise outcomes to a coherent set of item scores. This requirement motivates robust and computationally efficient probabilistic ranking models. The Bradley-Terry (BT) (Bradley & Terry, 1952) model formalizes this mapping by estimating item-specific scores from binary comparison outcomes. It is consequently a standard model class for aggregating pairwise judgments across application domains.

Despite this utility, the standard Bradley–Terry model is limited in practical human-evaluation settings. BT parameters can be estimated by iterative scaling methods (Ford Jr, 1957), gradient descent, or minorization maximization (MM) (Hunter, 2004). For the basic BT model, MM provides convergence to the global maximum, whereas gradient-based estimators, including procedures closely related to Elo rating, are common in practice but do not provide the same guarantee (Hunter, 2004). For extended BT models that include covariates or outcome structures such as home field advantage (Agresti, 2010), multiple comparisons (Plackett, 1975; Luce et al., 1959), or ties (Rao & Kupper, 1967), MM procedures may themselves converge only to local optima (Hunter, 2004). An additional modeling difficulty is rater heterogeneity: human participants may guess, rush, or lose focus, producing comparison outcomes with nonuniform reliability. If the aggregation model treats all comparisons as exchangeable and equally reliable, low-quality raters can bias the estimated item scores and reduce the stability of the induced ranking.

We propose a Bayesian Bradley–Terry variant that jointly estimates item quality and rater reliability. To the best of our knowledge, this is the first BT-based Bayesian formulation that explicitly parameterizes rater quality while also yielding closed-form EM updates. Whereas prior noisy-rater approaches commonly use gradient-based optimization without convergence guarantees, the proposed estimator has a stable latent-variable construction and interpretable item- and rater-level parameters. The Bayesian specification introduces regularization and represents epistemic uncertainty in the item scores. Rater-specific quality parameters encode the degree to which each participant’s judgments are consistent with the BT preference process, thereby modulating the contribution of each individual comparison. We derive an EM algorithm that estimates both parameter families efficiently and guarantees monotonic improvement of the observed-data likelihood. The resulting estimator preserves useful signal from partially reliable raters while attenuating comparisons from inattentive or inconsistent participants. Because rater quality and item uncertainty are modeled explicitly, the method produces rankings that are both more interpretable and more suitable for comparison across studies.

We evaluate the method on multiple human-evaluation datasets and introduce a newly collected *in-homogeneous rater quality (IHQ)* dataset¹. Across these datasets, the proposed estimator converges faster, is more robust to noisy comparisons, and produces more consistent rankings than standard BT and naive aggregation procedures. Beyond generative-model evaluation, the framework applies to any setting in which noisy pairwise comparisons must be transformed into reliable global rankings. Overall, the contribution supports more cost-efficient, interpretable, and reproducible human studies for evaluating AI-generated content.

2 RELATED WORK

Human-subject studies frequently require estimation of an ordering over items. A two-alternative choice task, rather than an absolute scoring task, can increase sensitivity to small perceptual differences and reduce response variability (Zerman et al., 2018; Wang et al., 2023). For this reason, many machine-learning leaderboards are built from pairwise-comparison data. In these protocols, raters select a preferred alternative from each presented pair, and the resulting binary outcomes are aggregated into a global ranking. Representative examples include Chatbot Arena (Chiang et al., 2024) and the CLIC image compression challenge (CLIC, 2025), both of which collect pairwise preferences and aggregate them with Bradley-Terry (BT) variants.

The Bradley-Terry model (Bradley & Terry, 1952), originally developed for ranking sports competitors, defines a probabilistic preference relation between pairs of items. Given item-specific skill parameters, the model specifies the probability that one item is preferred to another and thereby converts binary comparisons into an estimated ranking. This formulation has become a core statistical tool for games, consumer-preference analysis, and related ranking problems. A substantial literature extends the model to richer observation structures and application constraints. For example, the Plackett-Luce model generalizes BT from pairwise outcomes to rankings over multiple items (Plackett, 1975; Luce et al., 1959) by defining a distribution over permutations through products of successive BT probabilities (Luce et al., 1959). Other extensions incorporate context-specific factors, including home-field advantages (Agresti, 2010), ties (Rao & Kupper, 1967), and comparisons between groups rather than individual items (Huang et al., 2006).

¹The IHQ dataset will be released upon acceptance.

The parameters of the basic BT model are typically estimated by maximum likelihood estimation (MLE). Zermelo (1929) introduced an iterative procedure for computing these estimates, and this procedure remains widely used. Lange et al. (2000) later showed that the update is an instance of minorization-maximization (MM), in which a tractable surrogate objective is optimized at each iteration. Hunter (2004) extended MM methods to generalized BT models and identified conditions under which the iterates converge to the MLE. For the classical BT model, convexity implies convergence to the global optimum (Hunter, 2004). For extensions involving home-field advantage, multiple comparisons, or ties, however, the same class of algorithms may converge only to a local maximum (Hunter, 2004).

Bayesian BT formulations have been developed to incorporate prior information, regularize skill estimates, and quantify posterior uncertainty (Adams, 2005; Guiver & Snelson, 2009; Caron & Doucet, 2012). Caron & Doucet (2012) showed that MM updates can be interpreted as Expectation-Maximization (EM) procedures, which enables Bayesian inference through Gibbs sampling. This construction yields tractable complete-data likelihoods and convergence properties for the resulting Markov Chain Monte Carlo procedure. From the Bayesian perspective, posterior smoothing reduces sensitivity to sparse observations and supports uncertainty quantification for the inferred item skills.

Annotator-quality modeling addresses the restrictive assumption that every comparison has identical reliability. In crowdsourced studies, rater competence and attentiveness can vary substantially, motivating probabilistic models that include annotator-specific reliability parameters (Chen et al., 2013). Chen et al. (2013) proposed a Bayesian model that jointly estimates item quality and rater reliability through annotator-specific parameters. Their formulation assigns Gaussian priors to item and rater parameters and includes likelihood scaling factors that modulate each annotator’s influence. Posterior inference is performed with gradient-based optimization, which can be difficult in high-dimensional parameter spaces and may be sensitive to local optima. Moreover, such gradient-based procedures do not provide convergence guarantees. By contrast, the EM estimator proposed here provides guaranteed monotonic improvement of the likelihood. The Elo-based rater model developed by Google Research implements a simplified version of the same general idea. That model is widely used in the Challenge on Learned Image Compression (CLIC) (CLIC, 2025) and in related image-compression studies (Mentzer et al., 2020; Ballé et al., 2025).

To the best of our knowledge, no existing method combines a Bradley–Terry preference model, an explicit Bayesian rater-quality component, and closed-form EM updates. Unlike Chen et al. (2013), whose model uses Gaussian priors and gradient-based optimization without convergence guarantees, our formulation uses conjugate priors and an EM procedure with monotonic likelihood improvement. Relative to the Elo-style rater model commonly used in CLIC (CLIC, 2025), which is a simplified heuristic update scheme, the proposed method supplies a principled Bayesian estimator with posterior uncertainty and interpretable rater-quality parameters.

3 METHODOLOGY

3.1 BRADLEY-TERRY MODEL WITH RATER QUALITY

The task is to transform noisy pairwise-comparison observations into a reliable ranking over items, a standard estimation problem in machine learning and statistics. Let there be K items and R raters, with raters repeatedly comparing items in pairs. We denote the resulting comparison-outcome dataset by D . For any two items i and j , Bradley & Terry (1952) specify the pairwise preference probability as follows:

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j} \quad (1)$$

where $\lambda_k > 0$ denotes the positive skill parameter associated with item $k \in \{1, 2, \dots, K\}$. The model therefore provides an interpretable mapping from observed wins and losses to latent item scores. Its basic form, however, assumes that every comparison is generated with the same reliability. Human-evaluation data often violate this assumption because participants differ in expertise, attention, and diligence. These differences induce rater-dependent noise levels and can produce inconsistent judgments.

To model heterogeneous reliability, we introduce a rater-specific quality parameter $q_r \in [0, 1]$. With probability q_r , rater r is assumed to generate an informed judgment according to the Bradley-Terry preference probability. With probability $1 - q_r$, the same rater is assumed to respond randomly, equivalently selecting between the two items by a fair coin flip. This yields the following mixture probability for the event that rater r ranks item i above item j :

$$P(r \text{ ranks } i \text{ above } j) = q_r \left(\frac{\lambda_i}{\lambda_i + \lambda_j} \right) + (1 - q_r) \left(\frac{1}{2} \right) \quad (2)$$

The corresponding observed-data log-likelihood is:

$$\begin{aligned} \log P(D \mid \lambda, q) = & \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[w_{r,ij} \log \left(q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right. \\ & \left. + (n_{r,ij} - w_{r,ij}) \log \left(q_r \frac{\lambda_j}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2} \right) \right], \end{aligned} \quad (3)$$

where $n_{r,ij}$ is the number of comparisons of items i and j made by rater r , and $w_{r,ij}$ is the number of those comparisons in which rater r ranks item i above item j . This likelihood does not admit a closed-form maximum likelihood estimator for the joint parameter vector (λ, q) . Generic iterative methods such as gradient descent can be applied, but their behavior depends on tuning choices and does not provide a monotonic-likelihood guarantee. By contrast, an Expectation–Maximization (EM) procedure avoids learning-rate selection and guarantees monotonic increase of the observed-data likelihood with convergence to a stationary point (Dempster et al., 1977).

3.2 THURSTONIAN INTERPRETATION

Following Caron & Doucet (2012), we use a Thurstonian representation of the Bradley-Terry model (Diaconis, 1988). Under this representation, comparing items i and j is equivalent to a race in which the items are assigned random arrival times Y_i and Y_j . These arrival times are modeled as exponential random variables:

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad (4)$$

and the item with the smaller realized arrival time is observed as the winner. This construction recovers the standard Bradley-Terry preference probability:

$$P(i \text{ beats } j) = P(Y_i < Y_j) = \frac{\lambda_i}{\lambda_i + \lambda_j}. \quad (5)$$

For the EM algorithm, we augment the model with latent variables that make the complete-data likelihood tractable. We first define the indicator variable

$$A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r), \quad (6)$$

which indicates whether the c -th comparison of pair (i, j) by rater r was generated by the Bradley-Terry component rather than by the random-response component. Using these indicators, we define $Z_{r,ij}$ as the sum of the minimum arrival times over the $n_{r,ij}$ comparisons made by rater r :

$$Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)}). \quad (7)$$

Conditional on $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$, the latent sum $Z_{r,ij}$ has the Gamma distribution

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j), \quad (8)$$

where $\Gamma(\alpha, \beta)$ denotes the Gamma distribution with shape parameter α and inverse scale β . This Gamma latent-variable representation yields tractable EM updates while preserving the rater-specific quality component.

3.3 EXPECTATION-MAXIMIZATION UPDATES

The EM algorithm computes maximum a posteriori (MAP) estimates of the model parameters, λ and q , by alternating between posterior expectations over latent variables and maximization with respect to the parameters. In this model, the latent structure consists of the unobserved Bradley-Terry-component indicators and the arrival-time variables from the Thurstonian construction. The resulting sequence converges to a stationary point of the posterior objective.

We assign conjugate priors to the two parameter families. Each item skill λ_k has a Gamma prior, $\lambda_k \sim \Gamma(a, b)$, which is conjugate to the exponential arrival-time model. Each rater-quality parameter q_r has a Beta prior, $q_r \sim B(\alpha, \beta)$.

At each iteration, EM maximizes the expected complete-data log-posterior conditioned on the current parameter values (λ^*, q^*) . The procedure alternates between the following expectation and maximization steps until the convergence criterion is satisfied:

E-step: Expectation In the E-step, we compute the expected complete-data log-posterior, denoted by Q . This objective is the conditional expectation of the complete-data log-posterior over all latent variables, given the observed comparisons and the current parameter estimates. Formally,

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)]. \quad (9)$$

The complete-data log-likelihood ℓ_c decomposes into three factors:

$$\ell_c(\lambda, q; D, Z, A) = \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q). \quad (10)$$

M-step: Maximization The M-step updates the model parameters by maximizing Q . Because the E-step supplies the required posterior expectations, the original coupled optimization reduces to closed-form parameter updates. Both updates depend on the posterior probability that a comparison made by rater r was generated by the Bradley-Terry component. We denote this probability by γ ; it functions as a comparison-level weight determined by the agreement between the observed judgment and the current model prediction. It is given by:

$$\gamma_{r,ij}^{(t-1)} = \frac{q_r^{(t-1)} y_{ij}^{(t-1)}}{q_r^{(t-1)} y_{ij}^{(t-1)} + (1 - q_r^{(t-1)}) \frac{1}{2}}, \quad (11)$$

where $y_{ij}^{(t-1)} = \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}$ is the Bradley-Terry probability, under the previous iterate, that item i beats item j . Thus γ is the posterior responsibility of the informed-comparison component for the observed outcome. Larger values assign greater effective weight to the corresponding comparison.

The update for rater quality q_r is a regularized weighted average. The numerator aggregates the rater's effective informed comparisons, with each observed judgment weighted by its posterior responsibility γ , and adds the Beta-prior contribution. The denominator normalizes by the total number of comparisons and the prior mass. Consequently, a rater receives a larger quality estimate when their judgments are consistently explained by the Bradley-Terry component. The update is

$$q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}] + (\alpha - 1)}{n_r + \alpha + \beta - 2}, \quad (12)$$

where n_r is the total number of comparisons by rater r .

The update for item skill λ_i balances effective positive evidence against the total comparison exposure of the item. The numerator sums the posterior-weighted wins of item i across raters and adds the Gamma-prior contribution. The denominator aggregates the posterior-weighted pairwise exposure terms involving item i and adds the Gamma rate parameter. The resulting closed-form update is

$$\lambda_i^{(t)} = \frac{\sum_{r=1}^R \left[\sum_{j=1, j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} \right] + (a - 1)}{\sum_{j=1, j \neq i}^K \left[\frac{\sum_{r=1}^R [w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}]}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} \right] + b}. \quad (13)$$

Appendix B provides the full derivation of the Expectation Maximization updates.

4 EXPERIMENTAL RESULTS

We evaluate the proposed **Bayesian Bradley-Terry** model with rater **Quality (BBQ)** against two BT-based baselines: Bayesian Bradley-Terry (Bayes-BT) (Caron & Doucet, 2012) and a gradient-descent BT model with rater-quality parameters (Crowd-BT) (Chen et al., 2013). For Crowd-BT, we use the Google Research implementation (Google, 2025), which was also used by CLIC (2025) and Ballé et al. (2025). All experimental hyperparameters are specified in Section C.

The evaluation covers several Bradley-Terry (BT) variants on datasets from language-model preference evaluation (HUMAINE (Team, 2025), MT-Bench (Zheng et al., 2023)) and image-compression evaluation (WD (Ballé et al., 2025), HiFiC (Mentzer et al., 2020), ConHa (Aczel & Wattenhofer, 2024)), together with a newly collected inhomogeneous rater quality (IHQ) dataset derived from the CLIC2024 image test set (CLIC, 2025). To isolate the effect of rater reliability, we partition IHQ into screened and unscreened subsets. This design permits separate analysis of high-quality, low-quality, and mixed-rater comparison regimes. The IHQ collection protocol is described in Section E; dataset-level statistics for comparisons, raters, and items are reported in Section D and Table 2.

Real-world datasets do not provide independent ground-truth rankings. We therefore first validate the estimators on simulated datasets, where all methods recover the same ordering as the sample size increases. For crowdsourced datasets, we use the ranking obtained from the full dataset as an empirical reference ordering. We assess whether this reference is plausible by comparing the top-ranked item produced by each method. For every dataset except IHQ-unscreened, all three models identify the same top item. On IHQ-unscreened, Crowd-BT and Bayes-BT do not select the reference image as the highest-quality item, whereas BBQ does.

A reliable aggregation method should produce a similar ordering under repeated sampling of the study. We estimate this stability with bootstrap resampling, which quantifies variability in the induced rankings. The bootstrap protocol is specified in Section F. The WD and HiFiC datasets require additional caution because their original studies used active pair selection. HiFiC used a binary-search strategy, whereas WD selected pairs according to maximum information gain. Consequently, bootstrap estimates for these two datasets may not represent variability under independent random pair sampling.

We report two stability metrics. Top-1 agreement is the fraction of bootstrap samples whose best item matches the best item under the full dataset. Kendall’s Tau (τ) measures ordinal correlation between bootstrap and reference rankings (Kendall, 1938), with larger values indicating greater rank stability. Top-1 agreement emphasizes correctness of the leading item, whereas Kendall’s Tau evaluates the complete ordering.

4.1 PERFORMANCE ACROSS DATASETS

We compare the methods across datasets to characterize ranking accuracy and sampling stability. Table 1 reports Top-1 agreement and Kendall’s Tau. Datasets with more comparisons per item, including MT-Bench, WD, and HiFiC, generally produce more stable rankings. HUMAINE deviates from this pattern, indicating that comparison count alone is insufficient to explain performance; rater consistency and population diversity also affect aggregation quality.

Overall, BBQ provides the strongest stability profile across datasets. It identifies the top-performing item most frequently and is tied for best Top-1 agreement on three datasets. For the full ranking, it obtains the highest Kendall’s Tau on five of eight datasets and ranks second on the remaining three.

The three datasets on which BBQ ranks second in Kendall’s Tau are characterized by high-quality and comparatively homogeneous rater populations. On MT-Bench, all models recover the top item perfectly. The WD dataset from Ballé et al. (2025) contains only five raters, likely the paper’s authors, which suggests a carefully controlled evaluation process. In IHQ-screened, raters were explicitly filtered for quality. When rater reliability is already high and homogeneous, an explicit rater-quality parameter provides less additional information, reducing the empirical gap between BBQ and simpler BT variants.

The screened and unscreened IHQ subsets show the effect of heterogeneous rater quality directly. When low-quality raters are present, as in CLIC-all and CLIC-unscreened, BBQ substantially outperforms Crowd-BT and Bayes-BT. All three models achieve their highest Top-1 accuracy on the

Table 1: Performance of BT-based aggregation methods across datasets. The upper block reports Top-1 agreement [%], and the lower block reports Kendall’s Tau. BBQ identifies the top-performing item most frequently, achieves the strongest full-ranking recovery on more than half of the datasets, and ranks second on the remaining datasets.

	HUMAINE	MT-Bench	WD	HiFiC	ConHa	IHQ		
						all	scr.	unscr.
Top-1 Agreement [%]								
Crowd-BT	85.60	100.00	99.29	98.63	<u>66.59</u>	<u>85.31</u>	98.57	<u>33.15</u>
Bayes-BT	<u>97.30</u>	100.00	100.00	100.00	57.30	75.07	<u>98.90</u>	24.32
BBQ (ours)	97.50	100.00	100.00	100.00	77.12	99.32	99.80	61.92
Kendall’s Tau								
Crowd-BT	0.9385	0.9743	0.9279	<u>0.9366</u>	0.9180	0.9245	0.9171	0.8482
Bayes-BT	<u>0.9443</u>	0.9569	0.9459	0.9293	0.9110	0.9240	0.9116	<u>0.8507</u>
BBQ (ours)	0.9463	<u>0.9675</u>	<u>0.9359</u>	0.9525	0.9265	0.9270	<u>0.9132</u>	0.8563

screened subset, confirming that rater selection improves the measurement signal. Although Crowd-BT also includes rater-quality parameters, its performance decreases on the full dataset, plausibly because each crowdsourced rater contributes fewer than 40 comparisons and therefore provides limited evidence for reliable rater-quality estimation. BBQ, in contrast, preserves strong performance without screening and shows only a small reduction in Top-1 accuracy, indicating robustness to low-quality raters.

Uniformly high rater quality is difficult to guarantee in practice. Large-scale crowdsourcing introduces heterogeneous annotator behavior, screening procedures add cost, and subjective perceptual factors can affect even careful raters. In this regime, BBQ provides a principled mechanism for retaining information from partially reliable raters while suppressing the influence of noisy contributions.

4.2 SCALING WITH RATERS AND COMPARISONS

As indicated by Table 1, datasets with more comparisons per item generally yield more stable estimates. The comparison budget can increase through two mechanisms: adding raters or increasing the number of comparisons assigned to each rater.

Figure 1 isolates the effect of these two factors for the three BT variants. The left column varies the number of raters while holding per-rater comparisons at the maximum; the right column varies per-rater comparisons while holding the number of raters at the maximum. Top-1 accuracy shows clear model separation, whereas Kendall’s τ remains similar across methods. Crowd-BT fails to converge with only one or two raters, which illustrates the advantage of the EM algorithm and its convergence guarantee relative to gradient descent.

Crowd-BT and BBQ have similar behavior when either the number of raters or the number of comparisons per rater is very small. BBQ requires sufficient observations across both dimensions to separate rater-quality parameters effectively. As either dimension increases, BBQ increasingly improves over Bayes-BT. Bayes-BT underperforms overall, but it can exceed Crowd-BT when only a small number of raters or per-rater comparisons is available, because Crowd-BT cannot estimate rater quality reliably in such sparse regimes.

4.3 RATER QUALITY

Figure 2 relates each rater’s agreement with the final ranking to the quality parameter inferred by BBQ. The relationship is positively correlated: raters whose judgments align more closely with the consensus ranking receive larger inferred quality values. For the filtered dataset, the Pearson correlation is $r = 0.551$ across 50 raters. For the unfiltered dataset, the correlation is stronger, with $r = 0.724$ across 62 raters.

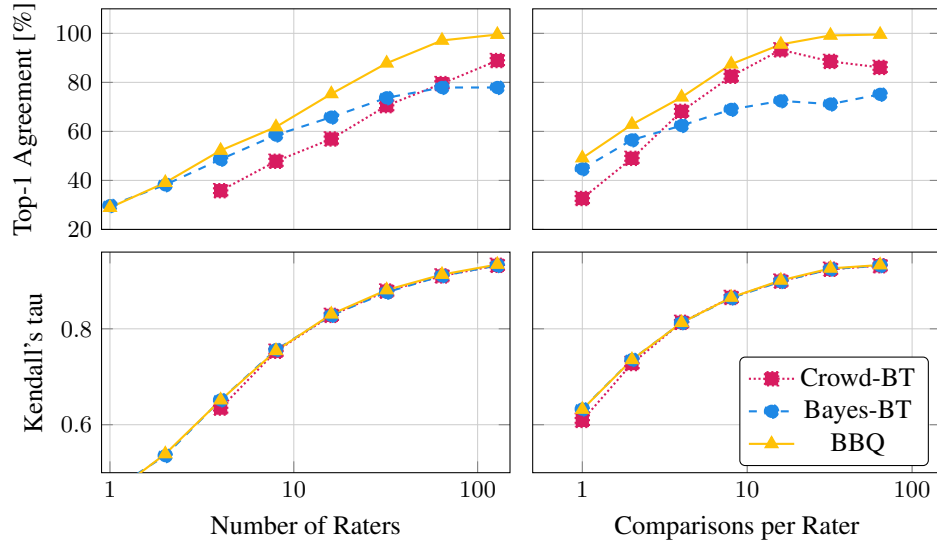


Figure 1: Scaling behavior of Bradley–Terry variants (Crowd-BT (Caron & Doucet, 2012), Bayes-BT (Chen et al., 2013), BBQ (ours)) on the IHQ-all dataset. **Left:** Performance as a function of the number of raters. **Right:** Performance as a function of the number of comparisons per rater. Both Top-1 agreement and Kendall’s τ increase as additional raters or comparisons are observed. Top-1 agreement separates the models more clearly, whereas Kendall’s τ remains comparatively similar across models. Crowd-BT does not converge with very few raters, illustrating the practical benefit of the EM-based estimator. Crowd-BT and BBQ are comparable in sparse regimes, but BBQ exceeds Bayes-BT as the number of raters or per-rater comparisons increases.

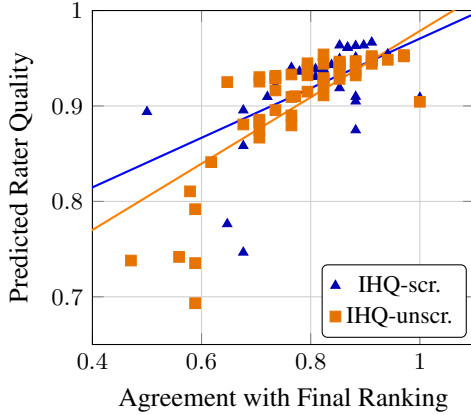


Figure 2: Rater-level relationship between agreement with the final ranking (x-axis) and predicted rater quality (y-axis) for the IHQ datasets. Each marker represents one rater. Triangles correspond to the filtered subset, and squares correspond to the unfiltered subset.

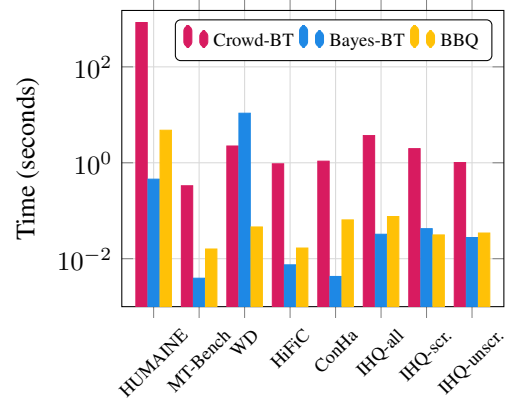


Figure 3: Average computation time in seconds (log-scale) for the three models on one bootstrapped sample from each of eight datasets. Crowd-BT can require substantial runtime on some datasets, whereas BBQ remains consistently fast across datasets.

As expected, the unscreened subset contains several raters with both low consensus agreement and low inferred quality. Some IHQ-unscreened raters fall below random-guessing agreement (50%), indicating systematic disagreement with the majority ranking. This pattern demonstrates why rater-quality modeling is necessary when aggregating pairwise comparisons from heterogeneous annotator pools. BBQ identifies these raters through lower quality estimates, thereby reducing their effective influence on the final ranking and limiting the noise they contribute.

4.4 COMPUTATIONAL EFFICIENCY

Figure 3 reports the average wall-clock time required by each method to process one bootstrapped sample for each dataset. These measurements characterize the computational feasibility of the estimators in repeated human-evaluation analyses. BBQ converges within a few seconds on every dataset. Crowd-BT is slower than BBQ across all datasets and is especially costly on comparison-dense datasets such as HUMAINE, where one run takes approximately 15 minutes. Bayes-BT is slow on WD, although it is slightly faster than BBQ on several smaller datasets. BBQ’s computational efficiency follows from the closed-form EM updates, which support rapid convergence even at larger dataset scales.

This runtime comparison is conservative with respect to BBQ: Crowd-BT uses a highly optimized C implementation (Kernighan & Ritchie, 1988), whereas BBQ is implemented in plain NumPy without targeted optimization. A compiled or more aggressively vectorized BBQ implementation would likely reduce runtime further. Thus, the method is not only statistically robust and stable but also practical for large-scale preference studies and for workflows that require repeated bootstrapping. The combination of accuracy, stability, and computational efficiency supports deployment in real evaluation pipelines.

4.5 CONFIDENCE INTERVALS

We evaluate uncertainty calibration for Crowd-BT, Bayes-BT, and BBQ in item-ranking inference. The controlled simulation uses pairwise comparisons between equally skilled items under H_0 . Each model estimates item skills and constructs uncertainty intervals for testing significant differences (Section G). The empirical Type I error rates are reported in Figure 6. At a 99% confidence level, the nominal error rate is 1%. Crowd-BT and BBQ closely match this target, with slightly lower observed rates, whereas Bayes-BT is more conservative at $\sim 0.1\%$. Thus, Crowd-BT and BBQ provide well-calibrated uncertainty estimates for this test, while Bayes-BT produces overly conservative intervals.

5 LIMITATIONS

Although BBQ explicitly models rater quality, the benefit of this component decreases when all raters are uniformly reliable. In homogeneous high-quality datasets, the rater-quality parameters contain little variation to estimate and therefore provide limited additional information. Maintaining such uniformly reliable rater pools, however, typically requires costly recruitment, screening, and monitoring procedures that may not scale to large studies. This tradeoff implies that BBQ is most advantageous in realistic crowdsourced settings with heterogeneous annotator behavior, and less critical in tightly curated evaluations.

A second limitation concerns the observation model. The current BBQ formulation is restricted to pairwise comparisons, whereas many human-evaluation protocols collect scalar ratings, full or partial rankings, or multi-way choices. Extending the likelihood and latent-variable construction to these feedback types would increase the scope of the framework.

Methodologically, the empirical analysis relies on bootstrap-based stability estimates. Although this procedure provides a practical measure of ranking variability, additional large-scale studies would further validate robustness across data-collection regimes. The model also assumes conditional independence across comparisons and does not include contextual, presentation-order, or fatigue effects, all of which may influence human judgments in practice.

Finally, although BBQ is computationally efficient in the evaluated settings, extremely large item or rater populations may still require optimized, vectorized, or compiled implementations.

6 CONCLUSION

We introduced BBQ, a Bayesian Bradley-Terry model for joint estimation of item quality and rater reliability. The central technical contribution is an Expectation-Maximization (EM) algorithm that estimates item skills and individual rater-quality parameters in a single probabilistic framework, thereby down-weighting or removing the effective influence of unreliable participants. Because

rater reliability is represented explicitly, the resulting rankings are more stable and accurate under heterogeneous annotation noise. The EM formulation provides rapid convergence and monotonic likelihood improvement, addressing important limitations of gradient-based alternatives.

Across diverse datasets, BBQ achieves high Top-1 agreement and strong Kendall’s Tau, indicating robust recovery of both the leading item and the broader ordering. The model is most beneficial in noisy or crowdsourced regimes, where raters may provide few comparisons and differ substantially in attentiveness or expertise. The empirical results therefore indicate that rater-quality modeling is especially important when evaluation data are heterogeneous or partially unreliable. The Bayesian component additionally supplies uncertainty estimates for item scores, supporting interpretable comparisons across studies. The uncertainty experiment shows that the resulting intervals are well calibrated for testing whether item differences are statistically significant. Finally, inferred rater quality correlates with agreement to the final ranking, supporting the interpretation of the quality parameter as a measure of evaluator reliability.

Overall, BBQ provides a practical, interpretable, and extensible aggregation method for noisy pairwise-comparison data. The framework contributes toward more cost-effective, statistically transparent, and reproducible human studies for evaluating AI-generated content.

REFERENCES

- Till Aczel and Roger Wattenhofer. Conditional hallucinations for image compression. *arXiv preprint arXiv:2410.19493*, 2024.
- Eldridge S Adams. Bayesian analysis of linear dominance hierarchies. *Animal Behaviour*, 69(5): 1191–1201, 2005.
- Alan Agresti. *Analysis of ordinal categorical data*. John Wiley & Sons, 2010.
- Jona Ballé, Luca Versari, Emilien Dupont, Hyunjik Kim, and Matthias Bauer. Good, cheap, and fast: Overfitted image compression with wasserstein distortion. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 23259–23268, 2025.
- Ralph Allan Bradley and Milton E Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.
- Francois Caron and Arnaud Doucet. Efficient bayesian inference for generalized bradley–terry models. *Journal of Computational and Graphical Statistics*, 21(1):174–196, 2012.
- Xi Chen, Paul N Bennett, Kevyn Collins-Thompson, and Eric Horvitz. Pairwise ranking aggregation in a crowdsourced setting. In *Proceedings of the sixth ACM international conference on Web search and data mining*, pp. 193–202, 2013.
- Wei-Lin Chiang, Lianmin Zheng, Ying Sheng, Anastasios Nikolas Angelopoulos, Tianle Li, Dacheng Li, Banghua Zhu, Hao Zhang, Michael Jordan, Joseph E Gonzalez, et al. Chatbot arena: An open platform for evaluating llms by human preference. In *Forty-first International Conference on Machine Learning*, 2024.
- CLIC. Clic 2024: Challenge on learned image compression. <https://archive.compression.cc/2024/>, 2025. Accessed: 2025-09-12.
- Arthur P Dempster, Nan M Laird, and Donald B Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the royal statistical society: series B (methodological)*, 39(1): 1–22, 1977.
- Persi Diaconis. Group representations in probability and statistics. *Lecture notes-monograph series*, 11:i–192, 1988.
- Lester R Ford Jr. Solution of a ranking problem from binary comparisons. *The American Mathematical Monthly*, 64(8P2):28–33, 1957.
- Rafael C Gonzalez. *Digital image processing*. Pearson education india, 2009.
- Google. Google, 2025. URL <https://github.com/google-research/google-research>. Accessed: 2025-09-12.
- John Guiver and Edward Snelson. Bayesian inference for plackett-luce ranking models. In *proceedings of the 26th annual international conference on machine learning*, pp. 377–384, 2009.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.
- Tzu-Kuo Huang, Ruby C Weng, Chih-Jen Lin, and Greg Ridgeway. Generalized bradley-terry models and multi-class probability estimates. *Journal of Machine Learning Research*, 7(1), 2006.
- David R Hunter. Mm algorithms for generalized bradley-terry models. *The annals of statistics*, 32(1):384–406, 2004.
- Frederick Jelinek. *Statistical methods for speech recognition*. MIT press, 1998.
- Maurice G Kendall. A new measure of rank correlation. *Biometrika*, 30(1-2):81–93, 1938.
- Brian W. Kernighan and Dennis M. Ritchie. *The C Programming Language*. Prentice Hall, 1988.

594 Kenneth Lange, David R Hunter, and Ilsoon Yang. Optimization transfer using surrogate objective
595 functions. *Journal of computational and graphical statistics*, 9(1):1–20, 2000.

596

597 Mabyduck Ltd. Mabyduck, 2025. URL <https://www.mabyduck.com/>. Accessed: 2025-09-
598 12.

599 R Duncan Luce et al. *Individual choice behavior*, volume 4. Wiley New York, 1959.

600

601 Fabian Mentzer, George D Toderici, Michael Tschannen, and Eiríkur Agustsson. High-fidelity gener-
602 ative image compression. *Advances in neural information processing systems*, 33:11913–11924,
603 2020.

604 Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic
605 evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association
606 for Computational Linguistics*, pp. 311–318, 2002.

607

608 Robin L Plackett. The analysis of permutations. *Journal of the Royal Statistical Society Series C:
609 Applied Statistics*, 24(2):193–202, 1975.

610 Pejaver V Rao and Lawrence L Kupper. Ties in paired-comparison experiments: A generalization
611 of the bradley-terry model. *Journal of the American Statistical Association*, 62(317):194–204,
612 1967.

613

614 Prolific AI Team. Humaine: Human-ai interaction evaluation dataset, 2025. URL [https://
615 huggingface.co/datasets/ProlificAI/humaine-evaluation-dataset](https://huggingface.co/datasets/ProlificAI/humaine-evaluation-dataset).

616 Kexin Wang, Yunlong Zhao, Qianqian Dong, Tom Ko, and Mingxuan Wang. Mospc: Mos prediction
617 based on pairwise comparison. *arXiv preprint arXiv:2306.10493*, 2023.

618

619 Zhou Wang, Eero P Simoncelli, and Alan C Bovik. Multiscale structural similarity for image quality
620 assessment. In *The thirty-seventh asilomar conference on signals, systems & computers, 2003*,
621 volume 2, pp. 1398–1402. Ieee, 2003.

622 Emin Zeman, Vedad Hulusic, Giuseppe Valenzise, Rafał K Mantiuk, and Frédéric Dufaux. The
623 relation between mos and pairwise comparisons and the importance of cross-content comparisons.
624 *Electronic Imaging*, 30:1–6, 2018.

625

626 Ernst Zermelo. Die berechnung der turnier-ergebnisse als ein maximumproblem der wahrschein-
627 lichkeitsrechnung. *Mathematische Zeitschrift*, 29(1):436–460, 1929.

628 Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang,
629 Zi Lin, Zhuohan Li, Dacheng Li, Eric. P Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica.
630 Judging llm-as-a-judge with mt-bench and chatbot arena, 2023.

A USAGE OF LLMs

During manuscript preparation, large language models (LLMs) were used as auxiliary tools. ChatGPT, Claude, Gemini, and Grammarly supported spellchecking, phrasing refinement, and text condensation for clarity and readability. ChatGPT, Claude, and Cursor were also used for code analysis, code completion, and visualization generation within the development workflow. These tools served only as support mechanisms; the core research ideas, experimental design, and interpretation of results were developed independently by the authors.

B BAYESIAN BT WITH RATER QUALITY DERIVATION

We consider R raters comparing K items. Each rater r has a quality parameter q_r , so that with probability q_r the comparison is generated by the Bradley–Terry model and with probability $1 - q_r$ the response is random. The derivation below gives the corresponding EM (Dempster et al., 1977) updates for incorporating rater-specific reliabilities into the Bayesian estimation framework.

B.1 MODEL DEFINITION

We use the following notation:

- $\lambda = (\lambda_1, \dots, \lambda_K)$: skill parameters for the K items.
- $q = (q_1, \dots, q_R)$: quality parameters for the R raters.
- $Y_i \sim \mathcal{E}(\lambda_i)$: latent arrival time associated with item i .
- $A_{r,ij}^{(c)} \sim \text{Bernoulli}(q_r)$: indicator that the c -th comparison of pair (i, j) by rater r follows the Bradley–Terry model.
- $n_{r,ij}$: total number of comparisons of (i, j) by rater r .
- $n_{ij} = \sum_{r=1}^R n_{r,ij}$: total number of comparisons of (i, j) .
- $w_{r,ij}$: number of times rater r ranked i above j .
- $w_{ij} = \sum_{r=1}^R w_{r,ij}$: total number of times i was ranked above j .
- $m_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model comparisons of (i, j) by rater r .
- $m_{ij} = \sum_{r=1}^R m_{r,ij}$: total number of Bradley–Terry-model comparisons of (i, j) .
- $v_{r,ij} = \sum_{c=1}^{w_{r,ij}} A_{r,ij}^{(c)}$: number of Bradley–Terry-model wins of i over j by rater r .
- $v_{ij} = \sum_{r=1}^R v_{r,ij}$: total number of Bradley–Terry-model wins of i over j .
- $Z_{r,ij} = \sum_{c=1}^{n_{r,ij}} A_{r,ij}^{(c)} \min(Y_i^{(c)}, Y_j^{(c)})$: sum of minimal arrival times, with

$$Z_{r,ij} \mid m_{r,ij} \sim \Gamma(m_{r,ij}, \lambda_i + \lambda_j).$$

The Bradley–Terry component defines the pairwise preference probability as

$$P(i \text{ beats } j) = \frac{\lambda_i}{\lambda_i + \lambda_j}.$$

Including rater quality gives the mixture probability

$$P(r \text{ ranks } i \text{ above } j) = q_r \frac{\lambda_i}{\lambda_i + \lambda_j} + (1 - q_r) \frac{1}{2}.$$

The equivalent latent exponential representation is

$$Y_i \sim \mathcal{E}(\lambda_i), \quad Y_j \sim \mathcal{E}(\lambda_j), \quad P(i \text{ beats } j) = P(Y_i < Y_j).$$

B.2 COMPLETE-DATA LOG-LIKELIHOOD

The complete-data log-likelihood is decomposed as

$$\begin{aligned}\ell_c(\lambda, q; D, Z, A) &= \log P(D, Z, A \mid \lambda, q) \\ &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q)\end{aligned}$$

The contribution from $P(Z \mid A, \lambda)$ is

$$\begin{aligned}\log P(Z \mid D, A, \lambda, q) &= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} \right. \\ &\quad \left. + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right]\end{aligned}$$

The contribution from $P(A \mid q)$ is

$$\log P(A \mid D, \lambda, q) = \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right]$$

The observation likelihood term $P(D \mid \lambda, A)$ yields

$$\begin{aligned}\log P(D \mid \lambda, q) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K \left[v_{r,ij} \log \frac{\lambda_i}{\lambda_i + \lambda_j} + (w_{r,ij} - v_{r,ij}) \log \frac{1}{2} \right] \\ &= \sum_{r=1}^R \sum_{i=1}^K [v_{ir} \log \lambda_i] \\ &\quad - \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K [(v_{r,ij} + v_{r,ji}) \log(\lambda_i + \lambda_j) + (w_{r,ij} + w_{r,ji} - v_{r,ij} - v_{r,ji}) \log 2] \\ &= \sum_{i=1}^K [v_i \log \lambda_i] \\ &\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2]\end{aligned}$$

Combining these terms gives the complete-data log-likelihood

$$\begin{aligned}
\ell_c(\lambda, q; D, Z, A) &= \log P(Z \mid D, A, \lambda, q) + \log P(A \mid D, \lambda, q) + \log P(D \mid \lambda, q) \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{ij} \log(\lambda_i + \lambda_j) - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i] \\
&\quad - \sum_{i=1}^K \sum_{j=i+1}^K [m_{ij} \log(\lambda_i + \lambda_j) + (n_{ij} - m_{ij}) \log 2] \\
&= \sum_{i=1}^K \sum_{j=i+1}^K \left[(n_{ij} - m_{ij}) \log 2 - (\lambda_i + \lambda_j) z_{ij} + (m_{ij} - 1) \log z_{ij} - \log \Gamma(m_{ij}) \right] \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[m_{r,ij} \log q_r + (n_{r,ij} - m_{r,ij}) \log(1 - q_r) \right] \\
&\quad + \sum_{i=1}^K [v_i \log \lambda_i]
\end{aligned}$$

B.3 EXPECTATION STEP

We introduce conjugate priors: a Gamma distribution $\lambda_i \sim \Gamma(a, b)$ for each item i , and Beta $q_r \sim B(\alpha, \beta)$ for each rater r .

The Q -function is the expectation of the complete-data log-posterior:

$$Q(\lambda, q \mid \lambda^*, q^*) = \mathbb{E}_{A, Z \mid D, \lambda^*, q^*} [\ell_c(\lambda, q; D, Z, A) + \log P(\lambda) + \log P(q)].$$

Posterior probability of a Bradley-Terry-consistent annotation:

$$\begin{aligned}
P(A_{r,ij}^{(k)} = 1 \mid i \succ j) &= \frac{P(i \succ j \mid A = 1) P(A = 1)}{P(i \succ j \mid A = 1) P(A = 1) + P(i \succ j \mid A = 0) P(A = 0)} \\
&= \frac{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^*}{\left(\frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} \right) q_r^* + \frac{1}{2} (1 - q_r^*)} = \gamma_{r,ij}^*.
\end{aligned}$$

Expected sufficient statistics:

$$\begin{aligned}
\mathbb{E}[m_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* \\
\mathbb{E}[m_{ij}] &= \sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \\
\mathbb{E}[v_{r,ij}] &= w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_{ir}] &= \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[v_i] &= \sum_{r=1}^R \sum_{j:j \neq i}^K w_{r,ij} \gamma_{r,ij}^* \\
\mathbb{E}[z_{ij}] &= \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*}.
\end{aligned}$$

Expected complete-data log-posterior:

$$\begin{aligned}\mathbb{E}[\ell_c] = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \text{const.}\end{aligned}$$

Priors contribute:

$$\begin{aligned}\mathbb{E}[\log P(\lambda)] &= \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i], \\ \mathbb{E}[\log P(q)] &= \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)].\end{aligned}$$

Final Q -function:

$$\begin{aligned}Q(\lambda, q \mid \lambda^*, q^*) = & \sum_{i=1}^K \sum_{j=i+1}^K \left[-(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \right] \\ & + \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\ & \quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\ & + \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\ & + \sum_{i=1}^K [(a-1) \log \lambda_i - b \lambda_i] \\ & + \sum_{r=1}^R [(\alpha-1) \log q_r + (\beta-1) \log(1 - q_r)] \\ & + \text{const.}\end{aligned}$$

where

$$\gamma_{r,ij}^* = \frac{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*}}{q_r^* \frac{\lambda_i^*}{\lambda_i^* + \lambda_j^*} + (1 - q_r^*) \frac{1}{2}}.$$

B.4 M-STEP

The M-step maximizes the Q -function with respect to the parameters (λ, q) while holding the E-step expectations fixed.

Update for q_r The update for each rater-quality parameter q_r is obtained by maximizing Q with respect to q_r , including the Beta-prior term.

$$\begin{aligned}
Q(q \mid \lambda^*, q^*) &= \sum_{r=1}^R \sum_{i=1}^K \sum_{j=i+1}^K \left[(w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) \log q_r \right. \\
&\quad \left. + (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) \log(1 - q_r) \right] \\
&\quad + (\alpha - 1) \log q_r + (\beta - 1) \log(1 - q_r) + \text{const.} \\
\frac{\partial Q}{\partial q_r} &= \sum_{i=1}^K \sum_{j=i+1}^K \left[\frac{w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*}{q_r} - \frac{n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{1 - q_r} \right] \\
&\quad + \frac{\alpha - 1}{q_r} - \frac{\beta - 1}{1 - q_r} \\
&\stackrel{0}{=} \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{q_r} \\
&\quad - \frac{\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1)}{1 - q_r} \\
&\Rightarrow (1 - q_r) \left(\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1) \right) \\
&\Rightarrow q_r \left(\sum_{i=1}^K \sum_{j=i+1}^K (n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta - 1) \right) \\
&\Rightarrow q_r = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*) + (\alpha - 1)}{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^* + n_{r,ij} - (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)) + (\beta + \alpha - 2)} \\
&\Rightarrow q_r^{(t)} = \frac{\sum_{i=1}^K \sum_{j=i+1}^K (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)}) + (\alpha - 1)}{n_r + \beta + \alpha - 2}
\end{aligned}$$

where the final line gives the explicit iteration- t update.

Update for λ_i The update for each item skill λ_i is obtained by maximizing Q with respect to λ_i , including the Gamma-prior term.

$$\begin{aligned}
Q(\lambda \mid \lambda^*, q^*) &= \sum_{i=1}^K \sum_{j=i+1}^K -(\lambda_i + \lambda_j) \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} \\
&\quad + \sum_{r=1}^R \sum_{i=1}^K \sum_{\substack{j=1 \\ j \neq i}}^K w_{r,ij} \gamma_{r,ij}^* \log \lambda_i \\
&\quad + \sum_{i=1}^K [(a - 1) \log \lambda_i - b \lambda_i] + \text{const.}
\end{aligned}$$

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$$\begin{aligned}
\frac{\partial Q}{\partial \lambda_i} &= - \sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} + \sum_{r=1}^R \frac{\sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^*}{\lambda_i} + \frac{a-1}{\lambda_i} - b \\
0 &\stackrel{!}{=} \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^* + (a-1)}{\lambda_i} - \sum_{j \neq i}^R \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^* + w_{r,ji} \gamma_{r,ji}^*)}{\lambda_i^* + \lambda_j^*} - b \\
\Rightarrow \lambda_i^{(t)} &= \frac{\sum_{r=1}^R \sum_{j \neq i}^K w_{r,ij} \gamma_{r,ij}^{(t-1)} + (a-1)}{\sum_{j \neq i}^K \frac{\sum_{r=1}^R (w_{r,ij} \gamma_{r,ij}^{(t-1)} + w_{r,ji} \gamma_{r,ji}^{(t-1)})}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + b} \\
\gamma_{r,ij}^{(t-1)} &= \frac{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}}}{q_r^{(t-1)} \frac{\lambda_i^{(t-1)}}{\lambda_i^{(t-1)} + \lambda_j^{(t-1)}} + (1 - q_r^{(t-1)}) \frac{1}{2}}
\end{aligned}$$

where the final line gives the explicit iteration- t update.

C HYPERPARAMETERS

For Crowd-BT, we use the implementation of Google (2025) with its default hyperparameters.

For Bayes-BT and BBQ, the hyperparameters consist only of prior-distribution parameters and stopping thresholds. For both models, convergence is declared when no ELO score changes by more than 1 between consecutive iterations. The ELO score is computed from the skill parameter λ as:

$$\text{ELO} = \log(\text{skill}) \cdot \text{ELO_SCALE_FACTOR}. \quad (14)$$

where ELO_SCALE_FACTOR is set to 400.

For the skill parameters in both Bayes-BT and BBQ, we use a Gamma prior with shape 5 and rate 0.1. For BBQ, the Beta prior on rater quality uses $\alpha = 10$ and $\beta = 2$.

D DATASETS

We evaluate the models on human-preference datasets spanning language and image domains. Table 2 summarizes the dataset statistics.

For natural-language evaluation, we use HUMAINE (Team, 2025), a large-scale benchmark with more than 100k comparisons, and MT-Bench (Zheng et al., 2023), which contains crowd-worker model-comparison judgments for multi-turn dialogues.

For image-compression evaluation, we use WD (Ballé et al., 2025), a dense expert-labeled dataset with thousands of comparisons per rater; HiFiC (Mentzer et al., 2020), which evaluates learned image codecs; and ConHa (Aczel & Wattenhofer, 2024), which focuses on conditional generative models. These datasets vary in scale, rater count, and rater expertise, providing a broad robustness testbed.

We also introduce the inhomogeneous rater quality (**IHQ**) dataset, collected through a user study on CLIC2024 (CLIC, 2025) data using the *Mabyduck* platform. The dataset contains two-alternative-forced-choice (2AFC) judgments over 28 generative image-compression models. To analyze the effect of rater quality, we provide two subsets: (i) **screened**, containing raters who passed pre-screening checks for attention and display quality, and (ii) **unscreened**, containing all raters. The screened subset approximates a higher-reliability annotator pool, whereas the unscreened subset represents the noisier conditions typical of large-scale human evaluation.

This dataset collection supports evaluation on both large, relatively clean benchmarks and smaller, noisier regimes in which rater-reliability modeling is expected to be critical.

Table 2: Summary of datasets used in the experiments. The IHQ dataset was collected on the Mabyduck platform and includes both 2AFC and 3AFC settings. For IHQ, screened subsets contain only raters who passed pre-screening tests for attention and display quality, whereas unscreened subsets contain all raters.

dataset	# comparisons	# raters	# models	comp/rater	comp/model
HUMAINE (Team, 2025)	105,220	1,977	27	53.2	3897.0
MT-Bench (Zheng et al., 2023)	3,355	65	6	51.6	559.2
WD (Ballé et al., 2025)	16,659	5	30	3331.8	555.3
HiFiC (Mentzer et al., 2020)	5,220	20	9	261.0	580.0
ConHa (Aczel & Wattenhofer, 2024)	1,531	40	8	38.3	191.4
IHQ-all	4,074	112	28	36.4	145.5
IHQ-screened	2,012	50	28	40.2	71.9
IHQ-unscreened	2,062	62	28	33.3	73.6

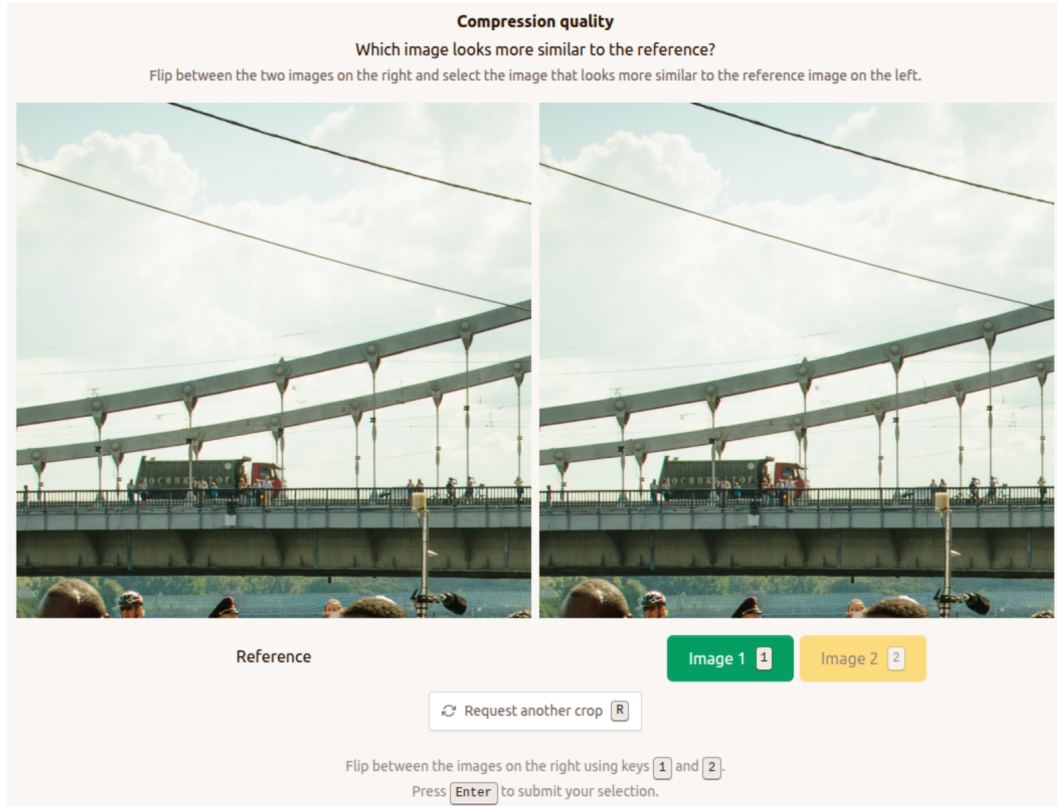


Figure 4: Screenshot of the Mabyduck user-study platform used to collect pairwise comparisons. The reference image appears on the left, and the rater selects one of two compressed images on the right.

E USER STUDY PLATFORM

All pairwise comparisons for the CLIC2024 (CLIC, 2025) dataset were collected with the Mabyduck platform (Ltd., 2025). Figure 4 shows the evaluation interface. The prompt asks: “Which image looks more similar to the reference image?” Ties are disallowed, and comparison pairs were sampled uniformly at random.

To analyze rater quality, we divide IHQ into screened and unscreened subsets. Screened raters passed pre-tests for color vision, display contrast, and sensitivity to subtle differences, and they

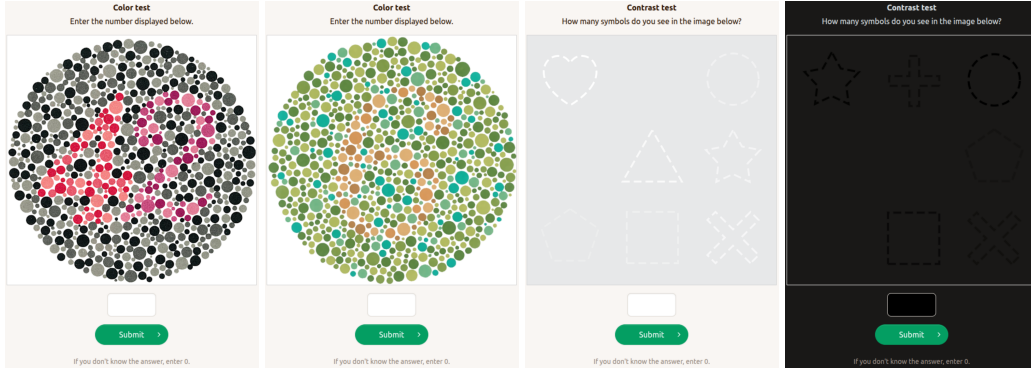


Figure 5: Four example images from the rater pre-screening procedure. The first two are color-blindness tests in which raters identify the number displayed in each pattern. The last two are shape-detection tests for sensitivity to low-contrast objects: one light gray shape on a white background and one dark gray shape on a black background.

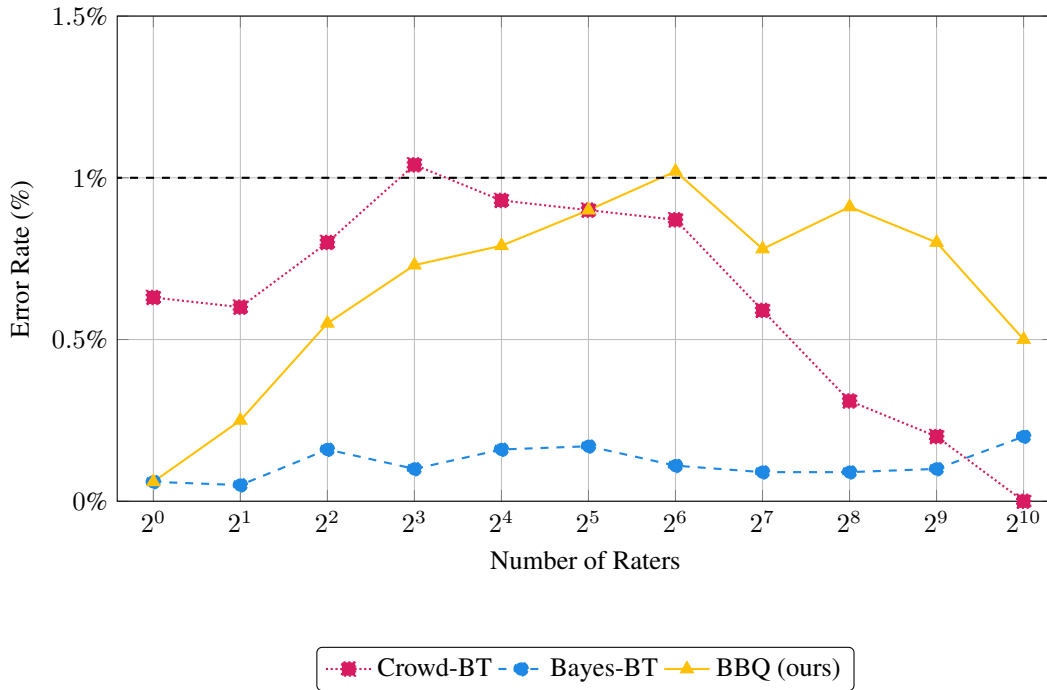


Figure 6: Type I error rate as a function of the number of raters. The dashed line denotes the nominal 1% error rate.

completed training comparisons that required selecting the higher-quality image to reinforce the evaluation criterion.

Figure 5 shows four examples from the pre-screening procedure. The first two are standard color-blindness tests requiring raters to identify the number in each pattern. The final two are shape-detection tests that measure sensitivity to low-contrast objects: one contains a light gray shape on a white background, and the other contains a dark gray shape on a black background. These tests restrict the screened subset to raters with sufficient visual capability for the task.

F BOOTSTRAPPING DETAILS

Because BBQ and Crowd-BT both include rater-quality parameters, the bootstrap resamples raters rather than individual comparisons. This procedure preserves each rater’s empirical comparison distribution and yields a stability estimate aligned with the model structure. For all datasets except HUMAINE, we draw 10,000 bootstrap resamples of raters. For HUMAINE, where one Crowd-BT bootstrap iteration takes approximately 15 minutes as discussed in Section 4.4, we use 1,000 resamples to control computational cost.

G UNCERTAINTY ESTIMATION DETAILS

Beyond estimating an ordering, the aggregation model should quantify uncertainty so that observed item differences can be assessed for statistical significance.

Bayesian Bradley-Terry (Bayes-BT) and Bayesian Bradley-Terry with Quality (BBQ) quantify uncertainty through the posterior distribution over item skills. Each skill parameter has a Gamma prior that is updated by the observed pairwise comparisons. Posterior credible intervals are then transformed to the Elo scale to support item-level comparison.

The classical Crowd-BT model is non-Bayesian, so uncertainty is estimated with a frequentist approximation. Specifically, we use a second-order Taylor expansion around the maximum likelihood estimate. Inverting the Hessian of the log-likelihood yields a covariance matrix whose diagonal entries provide item-level variances. These variances are converted to 99% confidence intervals via:

$$p_{99} = \sqrt{\text{diag}(\text{covariance})} \times k_{\text{Erfc}0.01} \times \sqrt{2} \approx \sqrt{\text{diag}(\text{covariance})} \times 3.29,$$

where the multiplicative constants map the standard deviation to the 99% confidence level. Sharper likelihood curvature yields narrower intervals and therefore higher certainty, whereas flatter curvature yields wider intervals and greater uncertainty.

To evaluate uncertainty calibration, we conduct a controlled simulation. Each trial contains two items with equal skill, corresponding to a 50–50 win probability, and pairwise outcomes are simulated by coin flips. For each trial, we simulate multiple raters and apply each model to estimate Elo scores and associated uncertainty intervals.

Formally, the null hypothesis H_0 states that the two items have equal strength. For each trial, we compute the 99% confidence or credible interval for each item-skill estimate. A model incorrectly rejects H_0 when the intervals do not overlap, thereby indicating a statistically significant item difference when no true difference exists. The primary metric is the frequency of these incorrect rejections, i.e., the observed Type I error rate.

We run 10,000 trials with 50 comparisons per rater and vary the number of raters to measure how calibration changes with data volume. This protocol estimates the empirical Type I error rate for each model and compares it with the 1% nominal rate implied by 99% intervals. As shown in Figure 6, Crowd-BT and BBQ produce Type I error rates close to the expected 1%, indicating calibrated uncertainty estimates. Bayes-BT is more conservative, consistently producing lower-than-nominal error rates, which indicates that its credible intervals are less suitable for this significance-testing procedure.